

Functional dependency exercises

1. Consider the relational schema $R(A, B, C, D, E, F, G)$ and set F of functional dependencies:

$F = \{E \rightarrow C, G \rightarrow AD, B \rightarrow E, C \rightarrow BF\}$.

Find E^+ , BC^+ , BG^+ , AC^+ .

2. Let $F1 = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD\}$ and $F2 = \{A \rightarrow CD, E \rightarrow AH\}$.

Are $F1$ and $F2$ equivalent?

3. Consider two sets of FDs as $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$ and $G = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$

Are F and G equivalent?

4. Find the minimal cover of the set of functional dependencies given:

$\{A \rightarrow C, AB \rightarrow C, C \rightarrow DI, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C\}$

5. Minimize $\{A \rightarrow C, AC \rightarrow D, E \rightarrow H, E \rightarrow AD\}$

6. Given a set of functional dependencies $F = \{AB \rightarrow C, AC \rightarrow D, B \rightarrow E\}$

Fill in the table below all reasonable data that reflects the information contained in F

A (integer)	B (String)	C (Integer)	D (Real)	E (Character)

7. Given this table

DID	Dname	EID	Ename	PID	Pname	Btime
10	Finance	1	Huey	27	Alpha	4.5
10	Finance	5	Dewey	25	Beta	3
10	Finance	11	Louie	22	Gamma	7
14	R&D	2	Jack	26	Pail	8
14	R&D	4	Jill	21	Hill	9

Find all possible functional dependencies from the data of the table.

8. Find the minimal cover of

$F = \{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, D \rightarrow E, D \rightarrow G, BE \rightarrow C, CG \rightarrow B, CG \rightarrow D, CE \rightarrow A, CE \rightarrow G\}$